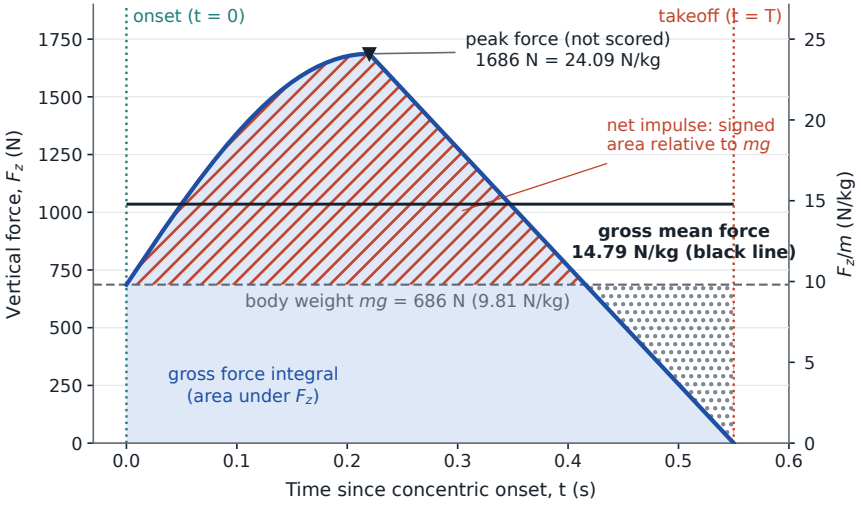
a **Concentric-phase force–time curve (analytical, not athlete data)**



b **Three distinct quantities**

Gross mean concentric force

$$Y_F = \frac{1}{mT}\int_0^T F_z\,dt$$

Basis of the scored force target

14.79 N/kg

Peak force

$$\max F_z / m$$

Not the scored target

24.09 N/kg

Gravity-adjusted net impulse

$$Y_J = \frac{1}{m}\int_0^T (F_z - mg)\,dt$$

Separate scored outcome

2.74 m/s

Illustrative: m = 70 kg, g = 9.80665 m s⁻², A = 1000 N, t_p = 0.22 s, T = 0.55 s.
t ≤ t_p: F = mg + A sin(πt / 2t_p)
t > t_p: F = (mg + A)[1 − (t − t_p)/(T − t_p)]

c **Assessment chronology relative to the index exposure (native contract and public synthetic task are separate)**

Assessment = mean of 3 valid trials · baseline = mean of the 2 latest valid assessments · no athlete data plotted

